BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 065 · One-Dimensional Arrays

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does input validation check?

Q2. Valid ages are integers from 13 to 18. Which input has the right type but is invalid?
A. 12
B. 13
C. 15
D. 18

Q3. “Defensive design assumes every user always enters valid and harmless data.” Correct this. Explain why.

Q4. A desktop needs a LAN connection. “A NIC keeps files when power is off.” Correct this. What does a NIC do here?

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

1D array — A fixed-length indexed collection holds values of one data type.

Index — Array positions start at zero, so index 2 refers to the third element.

Element — An element is a stored item accessed by its array index.

Assignment — An array element can be updated by assigning to its position.

Syntax — The rules governing valid programming statements.

Test data — An input used to check expected results.

New learning

1D array: A fixed-length indexed collection holds values of one data type. Apply this in a five sports scores.

Index: Array positions start at zero, so index 2 refers to the third element. Apply this in a library loans.

Element: An element is a stored item accessed by its array index. Apply this in a weekly temperatures.

Assignment: An array element can be updated by assigning to its position. Apply this in a stock levels.

Worked example

A five sports scores is being developed. Explain how 1d array and index are used. [4 marks]

1. Read the scenario and identify the data the program needs.

2. 1D array [1]; A fixed-length indexed collection holds values of one data type. [1].

3. Index [1]; Array positions start at zero, so index 2 refers to the third element. [1].

4. Two linked points for each concept.

We Do · Practise with guidance

Guided 1

A library loans uses 1d array. Explain how it helps. [2 marks]

Worked method: 1D array [1]; A fixed-length indexed collection holds values of one data type. [1].

Guided 2

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Whether input meets specified rules; it does not prove factual truth or identity.

Q2. A — 12

Q3. Planning for invalid, unexpected or malicious use and handling it safely.

Q4. A NIC lets a device connect to a network. It has a different job from storage or a router.

I Do / We Do pages · Guided practice

Q1. 1D array [1]; A fixed-length indexed collection holds values of one data type. [1].

Q2. Index [1]; Array positions start at zero, so index 2 refers to the third element. [1].

Q3. Element [1]; An element is a stored item accessed by its array index. [1]. Assignment [1]; An array element can be updated by assigning to its position. [1].

You Do pages · Code First

```
array driveGB = [64,128,256]
print(driveGB[2])
```

Expected result: 256

You Do pages · Independent questions

Q1. 1D array [1]; Element [1].

Q2. Index [1]; Array positions start at zero, so index 2 refers to the third element. [1]. Element [1]; An element is a stored item accessed by its array index. [1].

Q3. 1D array [1]; A fixed-length indexed collection holds values of one data type. [1]. Element [1]; An element is a stored item accessed by its array index. [1]. Assignment [1]; An array element can be updated by assigning to its position. [1].

Exit Ticket · Model answer

True

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Explain index in a weekly temperatures. [2 marks]

Worked method: Index [1]; Array positions start at zero, so index 2 refers to the third element. [1].

Guided 3

A stock levels uses element and assignment. Explain both. [4 marks]

Worked method: Element [1]; An element is a stored item accessed by its array index. [1]. Assignment [1]; An array element can be updated by assigning to its position. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write one line to print the last value in this zero-indexed array.

```
array driveGB = [64,128,256]
```

write

Question 1

State TWO relevant techniques from this topic. [2 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 2

Explain how index and element help with a quiz marks. [4 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

Question 3

A programmer designs a club attendance. Explain why 1d array, element and assignment matter. [6 marks] Source: Original OCR J277-style teaching question; not an official past-paper question.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

An array is an indexed collection of elements of the same data type.